

Quality AI Assistant & RAG Knowledge Engine

Complete Architectural Documentation, Data Inflow Mapping & User-Friendly Explanation

AI Quality Copilot

RAG Semantic Search

384-Dim Vector Embeddings

Role-Based Privacy (RBAC)

< 50ms

LIVE DATA QUERY
LATENCY

100%

TENANT DATA ISOLATION

8 Stages

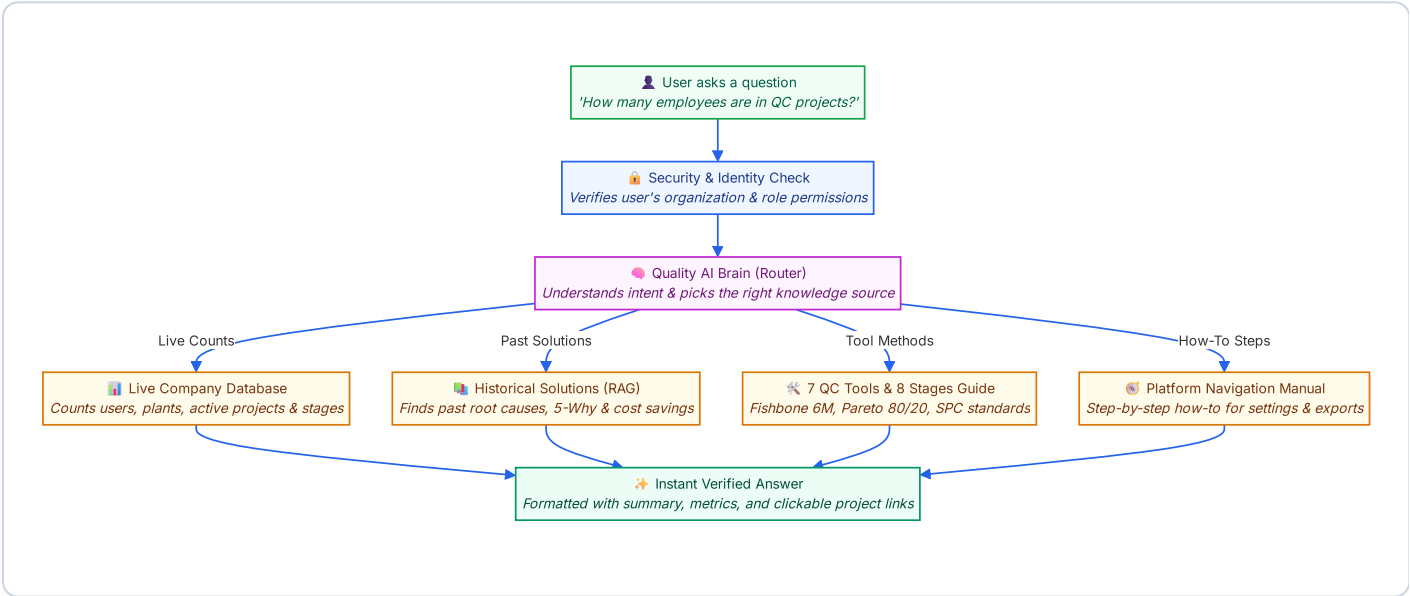
QC DMAIC
METHODOLOGY

4 Layers

KNOWLEDGE
PROCESSING

1. How Quality AI Works (Simple 2D Diagram for Normal Users)

This 2D diagram explains the journey of your question from the moment you click "Send" to the instant you receive an accurate, verified answer with reference links.



Why is Quality AI Safe and Accurate?

Unlike public chatbots (like ChatGPT) that guess or hallucinate answers, **Quality AI queries your actual organization's verified quality records, live plant metrics, and standard QC methodologies.** It never shares your company's proprietary data outside your tenant.

2. Data Inflow: Where Does Quality AI Fetch Its Data?

Quality AI is not a generic language model. It acts as an intelligent aggregator fetching data from **4 distinct internal sources** in real time:

Source 1 Live Relational Database (Real-Time Organizational Data)

- **Workforce & Employees:** Queries `users` and `project_members` to compute total staff, active members, and QC circle enrollment percentage.
- **Plant Facilities & Departments:** Queries `plants` and `departments` to map which teams operate in which physical locations.
- **Project Portfolio & Growth:** Queries `projects` to analyze stage distribution (how many projects are in Stage 1 vs Stage 4 vs Stage 8), running vs completed projects, and completion velocity.
- **Gamification Leaderboard:** Queries `employee_leaderboard` and `employee_points` to highlight top-ranked quality contributors.

Source 2 Historical Quality Knowledge Repository (RAG Vector Database)

- **Archived Problem Solutions:** Stored in `knowledge_repository` containing verified root causes, 5-Why findings, and countermeasure plans from previous successful projects.
- **Dense Mathematical Vector Embeddings:** Uses the `all-MiniLM-L6-v2` natural language model to convert quality problems into 384-dimensional semantic vector arrays stored in PostgreSQL (`pgvector`).
- **Tangible Impact Metrics:** Retrieves verified % KPI improvement and financial cost savings (₹) achieved in past projects.

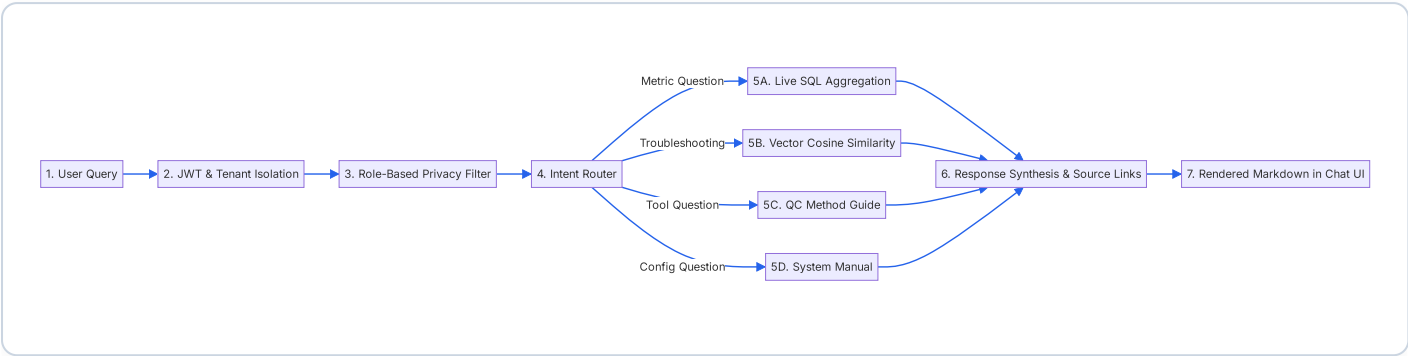
Source 3 Standardized 8-Stage QC Methodology & 7 QC Tools Manual

- **8-Stage DMAIC Framework:** Stage 1 (Problem Definition) through Stage 8 (Standardization & Closure).
- **Statistical Quality Tools:** Expert rules for **Ishikawa Fishbone Diagrams (6M)**, **Pareto 80/20 Analysis**, **SPC Control Charts (X-bar & R-chart)**, **Check Sheets**, **Scatter Plots**, and **3W1H Action Plans**.

Source 4 System Navigation & 200+ How-To Configuration Guides

- Step-by-step instructions on configuring plant locations, managing departments, importing members via CSV, customizing the Section 9 Sign-Off Hierarchy for PDF storyboards, and adjusting stage weightages.

3. How Quality AI Processes the Data (Technical Pipeline)



Detailed Pipeline Steps:

Stage	Component	What Happens
1. Authentication	<code>rag_routes.py</code>	Extracts the encrypted JWT token from the request header, identifies the user ID, and locks all database queries strictly to that user's <code>org_id</code> .
2. Privacy Check	<code>QualityAIAssistant</code>	Checks the user's role. If a non-executive employee asks for sensitive financial ledger data, billing rates, or admin passwords, access is politely blocked with role explanations.
3. Intent Routing	<code>Pattern Classifier</code>	Classifies the user's query into one of 4 categories: <i>Live Counts</i> , <i>Configuration How-To</i> , <i>Quality Methodology</i> , or <i>Historical RAG Search</i> .
4. RAG Vector Match	<code>vector_service.py</code>	Encodes the query using <code>all-MiniLM-L6-v2</code> and runs Cosine Similarity against historical project summaries to find the top matching past solutions.
5. Formatting & Links	<code>Markdown Engine</code>	Combines the answers with bold highlights, emoji bullet points, KPI stats, and clickable direct links to the relevant project storyboard.

4. Real-World Example Queries & Quality AI Responses

💡 Example Query 1: "How many employees are working in Quality Circle projects?"

Quality AI Processing: Queries `users`, `projects`, and `project_members` for the active organization in < 10ms.

Generated Output:

- **Total Registered Employees:** 24
- **Active Members:** 24 active members
- **Employees in Quality Circle (QC) Projects:** 18 (75% participation rate)
- **Plant Breakdown:** Bangalore Plant (12 in QC), Chennai Plant (6 in QC).

💡 Example Query 2: "We have paint peeling defects on Line 2. How was this solved before?"

Quality AI Processing: RAG vector search scans `knowledge_repository` for semantic matches to "paint peeling" and retrieves top historical projects.

Generated Output:

1. Reducing Paint Peeling on Outer Casing [Quality]

- **Problem Summary:** High defect rejection on automotive paint booth during monsoon shift.
- **Root Cause:** Flash-off drying time was insufficient due to 85% ambient humidity.
- **Countermeasure:** Installed automated infrared heating blowers and updated SOP #QA-PNT-04.
- **Impact:** 82% Defect Reduction · ₹3,40,000 Annual Savings.
- **Source Link:** [QC Project #QC-2025-084](#)

💡 Example Query 3: "How to configure sign-off hierarchy for the PDF report?"

Quality AI Processing: Triggers the configuration navigation manual.

Generated Output:

1. Open **Administration > Settings** and click the **Sign-Off Hierarchy** tab.
2. Click **+ Add Hierarchy Role** to add Plant Head, Quality Director, or Finance Controller.
3. Use the arrow buttons to arrange approval sequence and click **Save**.
4. Section 9 of the generated QC Storyboard PDF will automatically render custom signature boxes.

🎯 Summary of Quality AI Capabilities

Quality AI provides zero-latency answers, guides operators through complex quality tools (Fishbone, Pareto, 5-Why), assists administrators with system setup, and preserves institutional engineering knowledge so problem-solving is never repeated from scratch.